SURA Mid-Term Report

Hybrid Sensor Fused Indoor Localisation using KalmanNet

Abstract

Indoor positioning using hybrid smartphone sensors - Wi-Fi, magnetic field, and IMU - has attracted considerable research interest, but deep-learning models have consistently struggled to generalize in real-world deployments. This paper identifies three root causes: trajectory memorization, body-frame magnetic direction dependency, and non-causal look-ahead bias in bidirectional architectures. To address these, we propose a generalized, decoupled, and strictly causal Neural-Kalman fusion framework. An MLP environment model maps Wi-Fi RSSI fingerprints to a 2D probability heatmap; a 1D-CNN matches rotation-invariant magnetic sequences to pre-surveyed anomaly maps; both are fused with a PDR motion model through a dual-innovation KalmanNet with GRU-learned 2x2 gain matrices and sensor availability masks. The system achieves 0.47 m MAE under standard conditions and 1.07 m under degraded Wi-Fi, representing a 25.3% improvement over single-innovation baselines.

1. Introduction

Indoor localization is a foundational technology for location-based services, smart buildings, and emergency navigation. Because satellite signals attenuate severely indoors, systems must rely on ambient multi-modal smartphone sensors: primarily Wi-Fi Received Signal Strength Indication (RSSI), geomagnetic data, and Inertial Measurement Units (IMUs). These signals are complementary. Wi-Fi offers absolute but infrequent position fixes. IMU dead reckoning provides smooth, high-rate motion estimates but accumulates drift over time. The geomagnetic field presents a stable distortion pattern inside steel-reinforced buildings, providing a dense source of structural context.

Recent deep-learning approaches treat positioning as an end-to-end regression problem, using CNNs and bidirectional LSTMs to map sensor sequences directly to Cartesian coordinates. Despite promising benchmark results, three systematic problems prevent reliable real-world deployment.

Trajectory memorization. Models trained on a fixed set of walking paths learn the paths themselves rather than a generalizable map of the environment. Performance degrades sharply on unseen routes or when the same space is traversed in a different order.

Body-frame magnetic bias. Using raw Mag_x, Mag_y, Mag_z axes introduces a heading dependency: the same physical location appears different depending on which direction the user is facing. A model learning these raw features learns a direction-conditioned shortcut, not a spatial fingerprint.

Non-causal look-ahead bias. Bidirectional LSTM architectures process future observations to infer the current position. This violates causality and makes the model inapplicable to any real-time or online system regardless of its offline accuracy.

We propose a system architecture that resolves all three problems through explicit structural choices. The framework is evaluated on the MagWi Benchmark Dataset, focusing on the IT Engineering building of IIT Delhi.

2. Proposed System Architecture

The architecture functions as a modular learned Bayesian filter relying entirely on past and present observations, maintaining strict causality throughout. It comprises four components operating in a clear pipeline.

2.1 Problem formulation

The target environment is characterized by N uniquely identifiable Wi-Fi Access Points and a discrete reference grid of M surveyed spatial nodes. At each time step t the smartphone produces a Wi-Fi RSSI vector x_wifi in R^N and an IMU reading. The goal is to maintain a real-time 2D position estimate x_t using only observations up to and including time t.

2.2 Wi-Fi heatmap environment model (MLP)

An MLP maps the dense Wi-Fi RSSI fingerprint to a probability heatmap p over M discrete grid cells. Access Points not detected in a scan are assigned a constant floor value of -100 dBm. The network uses two fully connected layers (256 units, ReLU, 0.3 dropout) followed by a softmax output layer. The training target is a 2D Gaussian probability distribution centered on the true node coordinate with spatial standard deviation of 2.0 m, optimized using the KL divergence loss.

At inference, the continuous coordinate estimate z_wifi and associated measurement covariance R_hm are computed as the probability-weighted expectation (soft-argmax) over all grid cells. A concentrated heatmap yields a tightly bounded covariance; an ambiguous heatmap yields a large one. This uncertainty propagates directly into the Kalman fusion step.

2.3 Magnetic sequence model (1D-CNN)

A 1D-CNN processes a temporal window of T = 84 frames (5.0 seconds at 16.7 Hz) of rotation-invariant magnetic features: field magnitude |M|, vertical and horizontal projections, and the dip angle. This window size was selected via a sweep over 50, 84, 134, and 167 frames, balancing sufficient spatial variation for unambiguous matching against the risk of overfitting long topological routes.

The CNN outputs a coordinate estimate z_mag and a log-variance log(sigma^2), optimized with a heteroscedastic Negative Log-Likelihood loss that balances coordinate accuracy against self-reported confidence. The model automatically expresses lower confidence when the magnetic environment is ambiguous.

2.4 Causal Pedestrian Dead Reckoning (PDR)

The PDR system computes relative displacement at the full IMU rate. Step detection applies a first-order exponential high-pass filter (alpha = 0.98) to the acceleration magnitude, triggering on residuals exceeding 0.6 m/s^2 subject to a 0.3 s refractory period. The filter removes the static gravitational component (~9.81 m/s^2), isolating dynamic transients from foot impacts.

Each step produces a displacement vector u_t from a calibrated step length L_s, the real-time yaw heading from the device orientation sensor, and a constant angular offset aligning the device frame to the map coordinate frame. Step length is calibrated by matching the integrated step count to the ground-truth path length over training walks.

2.5 Dual-innovation KalmanNet fusion

Classical Extended Kalman Filters use fixed covariance assumptions that fail in variable multi-modal environments. We implement KalmanNet, replacing the analytical Kalman gain with a learned gain from a Gated Recurrent Unit (GRU). The prediction step follows standard dead reckoning: x_pred = x_{t-1} + u_t.

Two innovations are computed in parallel. The Wi-Fi innovation is the spatial residual between the MLP estimate and the predicted state. The magnetic innovation operates in the scalar anomaly domain: the observed anomaly is compared to the pre-surveyed map value, and the scalar mismatch is scaled by the spatial gradient of the anomaly field to produce a 2D spatial correction.

The GRU input is a 13-dimensional feature vector comprising the Wi-Fi innovation (2), scalar magnetic innovation (1), field gradient (2), temporal difference of consecutive Wi-Fi fixes (2), PDR control (2), previous state update (2), and binary availability masks for each modality (1+1). The GRU outputs two independent 2x2 gain matrices K_wifi and K_mag. The posterior update is: x_t = x_pred + m_wifi *

K_wifi * y_wifi + m_mag * K_mag * (y_mag * grad_A). The binary masks allow the filter to degrade gracefully when any sensor modality is unavailable.

3. Experimental Setup and Dataset

The system is evaluated on the MagWi Benchmark Dataset, using the IT Engineering building at IIT Delhi - a multi-floor structure with a complex pathway network suited for testing tracking resilience. The framework is building-agnostic and can be deployed on any structure following an initial reference grid survey.

3.1 Dataset augmentation

Continuous dataset recordings label only the starting node, lacking per-frame trajectory ground truth needed for rigorous temporal evaluation of the KalmanNet. Synthetic but physically realistic trajectories are generated using the following procedure.

Path generation. An epsilon-graph is constructed over the 168 surveyed nodes, connecting any two nodes separated by at most 1.6 m. Trajectories traverse shortest paths between randomized endpoint pairs through this topologically accurate corridor network.

Sensor simulation. Gait speed is sampled uniformly between 1.0 and 1.35 m/s. Simulated heading tracks the true geometric path tangent, subject to random-walk drift and 8.8 degree white noise variance matching the empirically measured gyroscope noise of the dataset.

This pipeline produced 250 trajectories for training and 60 independent trajectories for testing. Data from the Samsung Galaxy S9+ was held out entirely during training and reserved solely for evaluation.

3.2 Training details

All components were implemented in PyTorch. The MLP and CNN were optimized via Adam (beta_1 = 0.9, beta_2 = 0.999, weight decay 1e-4) with an initial learning rate of 1e-3, decayed by a ReduceLROnPlateau scheduler (patience 8, factor 0.5). The MLP trained for 80 epochs; the CNN for 60 epochs. The KalmanNet GRU trained for 150 epochs via Adam (lr = 2e-3, weight decay 1e-5) with MSE loss against ground-truth trajectories.

4. Results

4.1 Wi-Fi heatmap environment model

Table 1 documents the standalone performance of the Wi-Fi MLP prior to temporal fusion. The model achieves 1.43 m MAE on standard random splits and maintains 2.02 m MAE on entirely unseen smartphone hardware, demonstrating that the fingerprint representation generalizes across device types.

Configuration	Test device	MAE	Median	P90	Max
Random split	Mixed	1.43 m	1.12 m	2.65 m	7.84 m

Configuration	Test device	MAE	Median	P90	Max
Held-out device	Samsung S9+	2.02 m	1.68 m	3.74 m	9.12 m

Table 1. Wi-Fi heatmap environment model performance.

4.2 Dual-innovation KalmanNet fusion

Table 2 compares fusion configurations across 60 independent test walks under two signal regimes. The standard regime uses 1 Hz Wi-Fi fixes. The degraded regime restricts Wi-Fi to one reading every 5 seconds with a 40% AP dropout rate, simulating real-world signal outages in dense indoor environments.

Regime	Model configuration	Mean error	Median
Full Wi-Fi (1 Hz)	Wi-Fi-only KalmanNet	0.55 +/- 0.05 m	0.50 m
	Dual KalmanNet (+ Magnetic)	0.47 +/- 0.03 m	0.45 m
Degraded Wi-Fi (5 s gap, 40% dropout)	Wi-Fi-only KalmanNet	1.44 +/- 0.18 m	1.25 m
	Dual KalmanNet (+ Magnetic)	1.07 +/- 0.10 m	0.97 m

Table 2. Dual-innovation fusion comparison across 60 test walks.

Under full Wi-Fi conditions, adding continuous magnetic sequence data provides a 13.4% improvement in accuracy (0.55 m to 0.47 m MAE). The degraded regime reveals the true strength of the dual-innovation design. When Wi-Fi is sparse and incomplete, the single-innovation approach drifts significantly. The continuous 16.7 Hz magnetic updates act as a dense bridge, yielding a **25.3% reduction in MAE** from 1.44 m to 1.07 m.

4.3 Robustness and postural independence

Training data includes three device holding modes from the MagWi dataset: standard navigation hold, call listening (device near the ear), and natural arm swinging. The architecture is inherently robust to these variations because the Wi-Fi MLP operates on scalar RSSI values (orientation-independent) and the magnetic CNN processes rotation-invariant features. This postural agnosticism is critical for real-world deployment where users hold devices in unpredictable orientations.

5. Conclusion

This paper presented a generalized, decoupled Neural-Kalman fusion architecture for hybrid indoor positioning. Enforcing strict structural separation between spatial inference and temporal tracking avoids trajectory memorization and orientation-dependent bias. A 1D-CNN magnetic sequence matcher operates alongside an MLP Wi-Fi probability heatmap to provide robust spatial updates at two complementary timescales: dense 16.7 Hz magnetic anchors and periodic Wi-Fi fixes.

Fusing these through an extended KalmanNet enables context-dependent 2x2 matrix gains computed dynamically, with native resilience against missing sensor modalities via availability masks. In degraded signal scenarios - a common reality in physical deployments - the dual-innovation system delivers a 25.3%

improvement over single-innovation baselines.

Current configurations require building-specific training due to unique AP signature foundations. Future work will focus on extending the architecture to full three-dimensional, multi-floor positioning by integrating barometric pressure altitude data into the KalmanNet state vector.